

A

Stream-ablation design

 included  excluded

Ablation condition

	Self-pose geometry	Pairwise relations	Group-scene visual	Per-animal visual
Full multimodal reuse				
Animal visual + pose geometry, no group				
Pose geometry only				
Visual + self-pose, no pairwise geometry				
Per-animal visual only				
Visual descriptors only				

Controls held constant across rows

Frozen cache	32-frame windows	Train/val/test split	Matched seeds	Same decoder family within comparison
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